



TEAM DETAILS

Team Name:

TECHTONICS

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	AKSHAYA SRELEKA P S	III YEAR
2	Member 1	ASHWINKUMAR K	III YEAR
3	Member 2	DHAMARAI KANNAN A	III YEAR
4	Member 3	DHARSHAN S	III YEAR

COLLEGE NAME

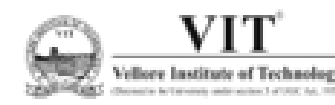
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PROBLEM STATEMENT

Drift-Sense: AI-Powered Navigation-Error Recovery for Wafer Inspection Tools

PROBLEM DESCRIPTION:

Given a high-magnification Reference Image and a 10× shrunk Search Image, the system must:

- Locate where the reference pattern appears inside the search image.
- Return the matching center coordinates (x, y) in pixels.
- Resolve multiple similar matches by selecting the candidate closest to the search-image center.

SOLUTION SUMMARY:

A physics-aware 5-phase cascade that recovers the correct SEM navigation site across magnification changes using multi-scale NCC, geometric verification, selective ML reranking, and sub-pixel Fourier refinement.

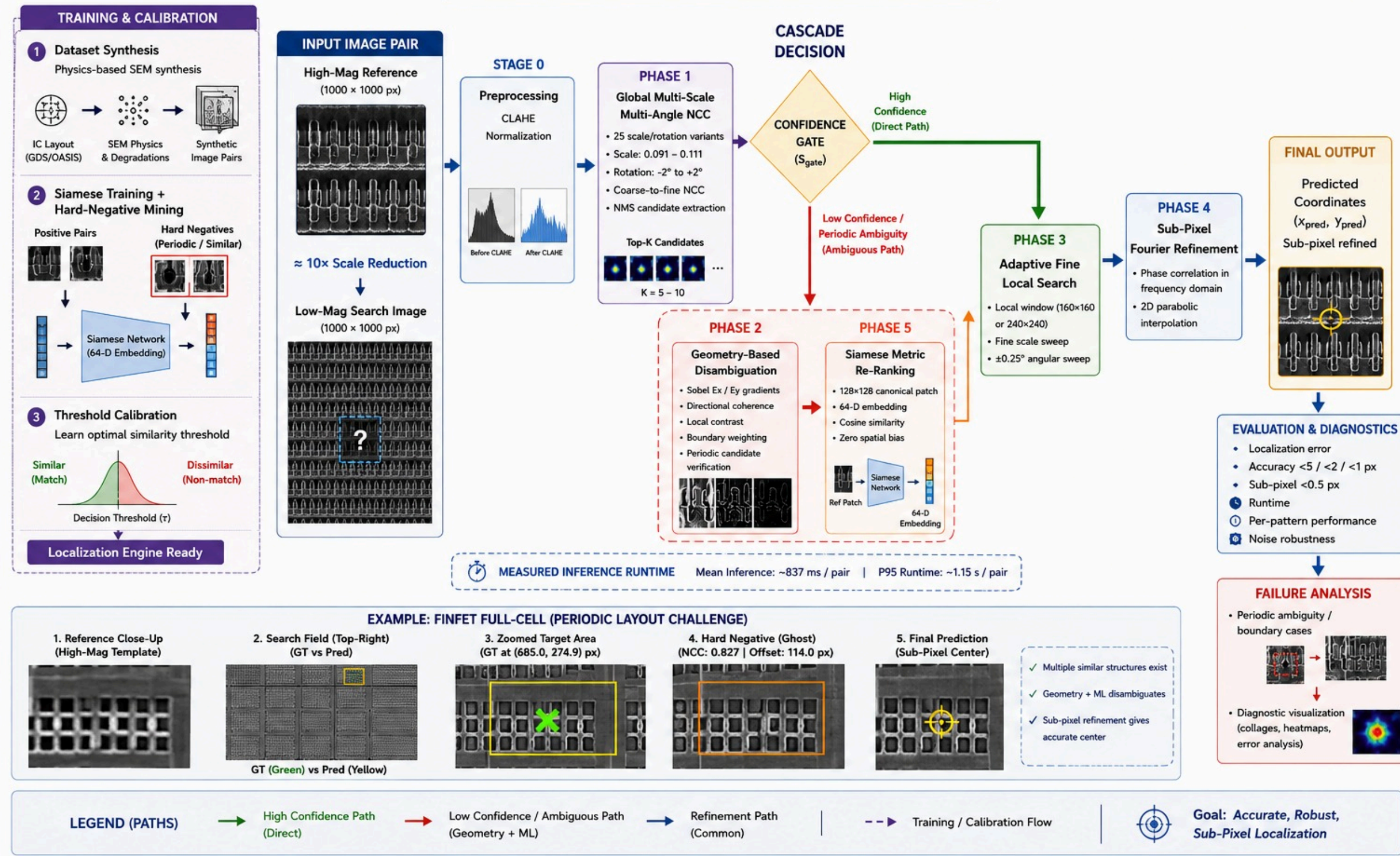
KEY RESULT:

98.33% Operational Accuracy (<5 px) | 86.67% High-Precision Accuracy (<1 px) | 0.7012 px Median Localization Error | 672.39 ms Mean Execution Latency

PROPOSED END-TO-END WORKFLOW

5-PHASE CASCADE LOCALIZATION PIPELINE

From Noisy Cross-Magnification SEM Images to Sub-Pixel Accurate Coordinates



INPUT CHALLENGES:

10× scale gap • SEM noise • scale/rotation variation • stage drift • periodic ambiguity

WHY CASCADE?

Fast NCC first → deeper processing only when ambiguity remains

DECISION LOGIC:

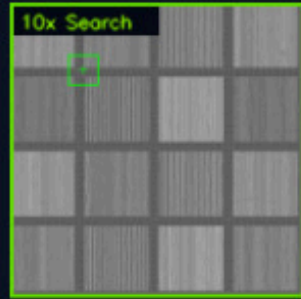
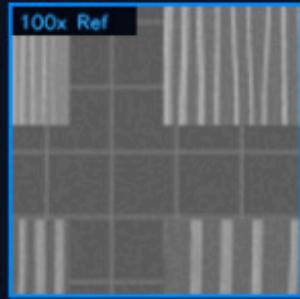
High confidence → Direct path
Low confidence / periodic ambiguity → Geometry → ML

OUTPUT:

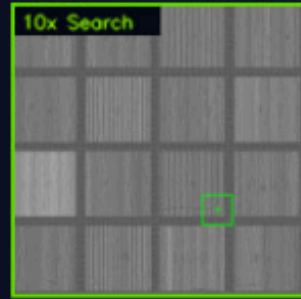
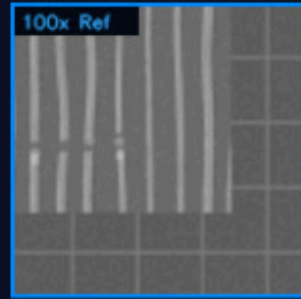
Predicted (x, y) → Fine local refinement → Sub-pixel coordinate

FINFET SYNTHETIC DATA DESIGN

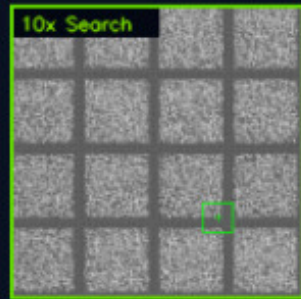
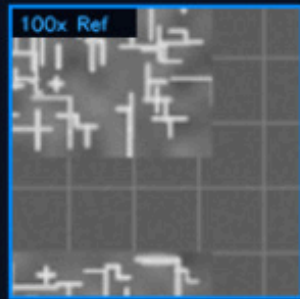
P1: FIN_ARRAY (SAQP Fin Grid)



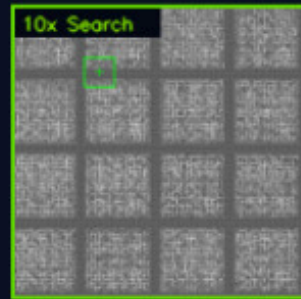
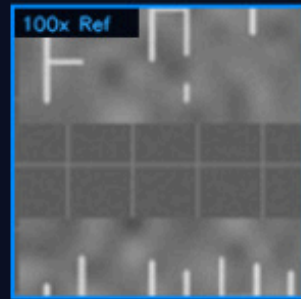
P2: FIN_CUT (EUV Cut Mask)



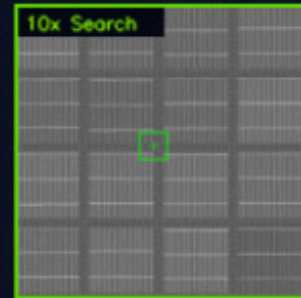
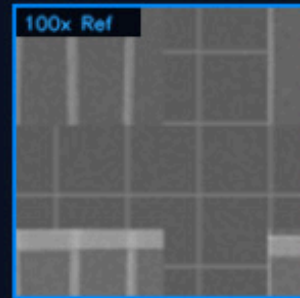
P5: LOCAL_INTERCONNECT (M0 Jogs)



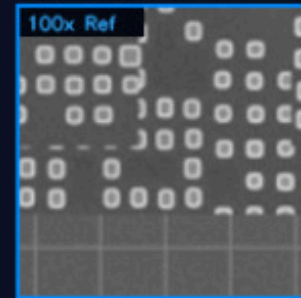
P6: METAL_ROUTING (M1/M2 Grid)



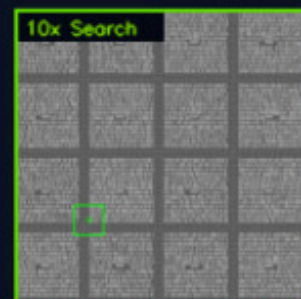
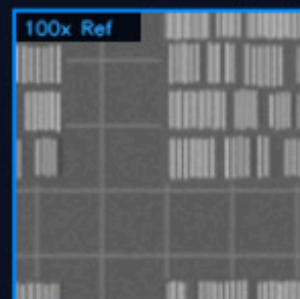
P3: FIN_GATE (3D Gate Matrix)



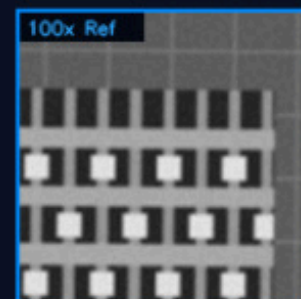
P4: CONTACT_ARRAY (BEOL CA/CB Vias)



P7: ACTIVE_CELL (Standard Cell)



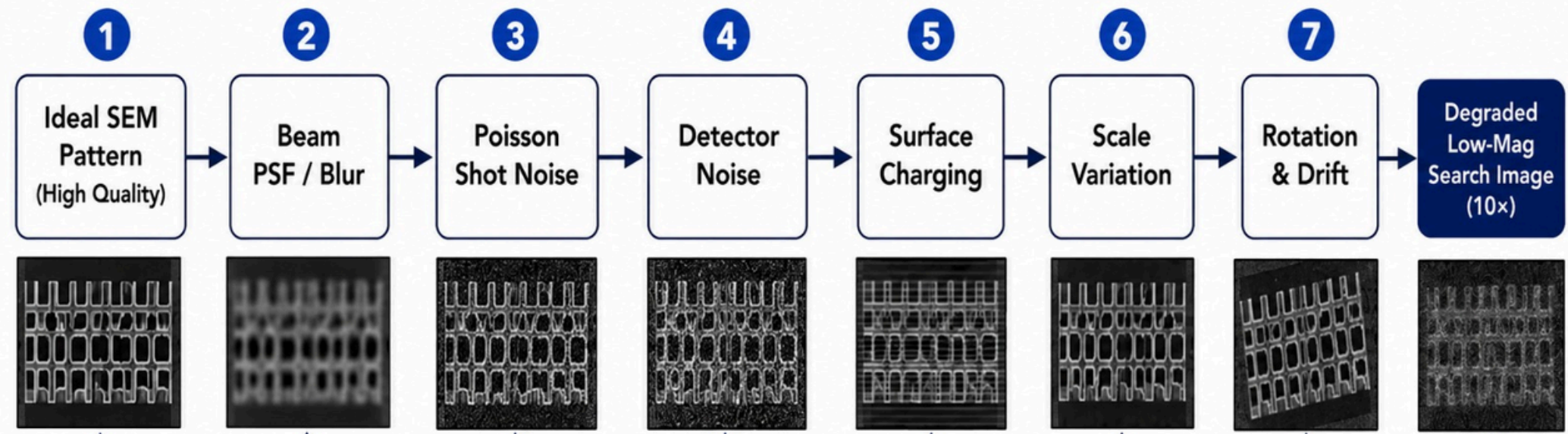
P8: FINFET_FULL_CELL (Full 3D FinFET)



CLASS	CLASS NAME	JUSTIFICATION
P1	FIN_ARRAY	Multi-scale and angular alignment under directional gradient structure
P2	FIN_CUT	Edge termination sensitivity and sub-pixel gap estimation
P3	FIN_GATE	2D spatial coherence and cross-junction tracking
P4	CONTACT_ARRAY	Periodic lattice ambiguity and pitch-hop disambiguation
P5	LOCAL_INTERCONNECT	Structural shape matching and local asymmetry
P6	METAL_ROUTING	Multi-layer contrast and routing-pattern matching
P7	ACTIVE_CELL	Asymmetric boundary localization
P8	FINFET_FULL_CELL	Complex periodicity and multi-layer cell ambiguity

SEM DEGRADATION & AUGMENTATION MODEL

DEGRADATION PIPELINE



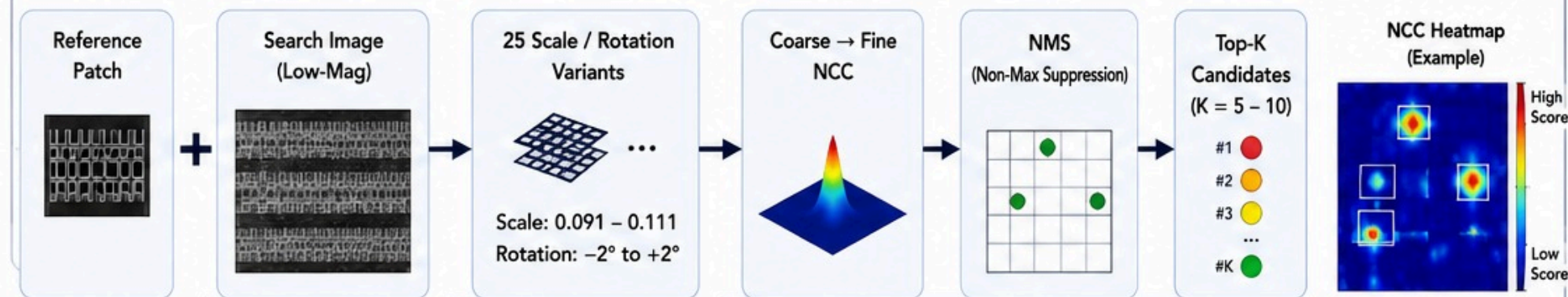
CITATIONS

- [1] Goldstein et al. (2018)
Scanning Electron Microscopy and X-Ray Microanalysis
<https://link.springer.com/book/10.1007/978-1-4939-6676-9>
- [2] Everhart & Thornley (1960)
Wide-band detector for micro-microampere low-energy electron currents
<https://doi.org/10.1088/0950-7671/37/7/307>
- [3] Li, Zhang & Ding (2013)
Monte Carlo Simulation of CD-SEM Images for Linewidth and Critical Dimension Metrology
<https://doi.org/10.1002/sca.21042>
- [4] Cazaux (1999)
Some considerations on the secondary electron emission, δ , from e^- irradiated insulators
<https://doi.org/10.1063/1.369239>

DEGRADATION MODEL	RANGE / PURPOSE IN EVALUATION	WHAT IT TESTS
Poisson Shot Noise	500–3500 e^-/pixel	Robustness to electron-counting noise and low beam dose
Detector Noise	$\sigma = 0.8\text{--}3.2$	Readout/detector noise
Beam Blur / PSF	$\sigma = 1.0\text{--}2.5 \text{ px}$	Localization under reduced spatial sharpness
Scale Variation	0.091–0.111	Cross-magnification scale mismatch
Rotation	$\pm 2.0^\circ$	Angular misalignment
Stage Drift	$\pm 11 \text{ px}$	Spatial positioning error
Surface Charging	Scanline intensity modulation	Charging-induced appearance variation

LOCALISATION METHOD & DECISION RULE

PHASE 1 – CANDIDATE GENERATION (Global NCC)



CONFIDENCE GATE

$$S_{\text{gate}} = 0.45S_{\text{top1}} + 0.35S_{\text{gap}} + 0.20S_{\text{PSR}}$$

HIGH CONFIDENCE (Direct Path)

✓ $S_{\text{gate}} \geq 0.65$ AND Peak Gap ≥ 0.075

LOW CONFIDENCE / PERIODIC AMBIGUITY

PHASE 2 – GEOMETRY-BASED DISAMBIGUATION

- Line/Fin spacing consistency
- Structural alignment score
- Periodicity verification

PHASE 5 – SIAMESE METRIC RE-RANKING

- Siamese CNN similarity score
- Re-rank Top-K candidates
- Select best candidate

COMMON REFINEMENT PATH (All Cases)

PHASE 3 – ADAPTIVE FINE LOCAL SEARCH

- Local window: 160×160 / 240×240
- Fine-scale sweep around best scale
- Angular sweep: ±0.25°

PHASE 4 – SUB-PIXEL FOURIER REFINEMENT

- Phase correlation (FFT domain)
- 2D parabolic interpolation
- Sub-pixel peak estimation

OUTPUT

Predicted Coordinates
(x_{pred} , y_{pred})



Observed in Benchmark:

NCC Direct 95.0%

Geometry Verified 0.8%

Siamese ML Reranked 4.2%

PHASE

PURPOSE SUMMARY

P1 – Global Multi-Scale NCC

Generate potential match locations using multi-scale, multi-angle NCC and return Top-K candidates.

P2 – Geometry-Based Disambiguation

Verify structural and periodic consistency using spacing, alignment, local contrast and periodic candidate verification to eliminate false matches.

P5 – Siamese Metric Re-Ranking

Use Siamese CNN similarity to re-rank difficult candidates and select the most likely match.

P3 – Adaptive Fine Local Search

Refine the selected candidate using a local search window, fine-scale sweep and small angular variations.

P4 – Sub-Pixel Fourier Refinement

Refine the final peak using frequency-domain phase correlation and 2D interpolation for accurate sub-pixel coordinates.

Output

Return the final predicted coordinates (x_{pred} , y_{pred}).

Decision Rule:

If $S_{\text{gate}} \geq 0.65$ AND Peak Gap ≥ 0.075 → accept the Top-1 NCC candidate and proceed directly to Phase 3 for fine localization.

Design Principle:

CLASSICAL FIRST → GEOMETRY WHEN AMBIGUOUS → ML AS SELECTIVE TIE-BREAKER → SUB-PIXEL REFINEMENT



IMPLEMENTATION & EXECUTION COMMANDS

1. ENVIRONMENT & DEPENDENCIES

- Python-based implementation
- OpenCV / NumPy for image processing
- PyTorch for Siamese metric learning
- SciPy for numerical and Fourier refinement
- Matplotlib / visualization utilities for diagnostics

TRAINING & CALIBRATION — OFFLINE

PHYSICS-BASED DATASET



SIAMESE TRAINING



HARD-NEGATIVE MINING



SIMILARITY THRESHOLD CALIBRATION



FROZEN LOCALIZATION MODEL

TRAINING IS PERFORMED OFFLINE; THE TRAINED MODEL IS SELECTIVELY INVOKED DURING INFERENCE.

COMMAND

FUNCTION

python
generate_dataset.py

Generate synthetic SEM benchmark pairs

python localize.py

Run the 5-phase localization engine

python
evaluate_predictions.py

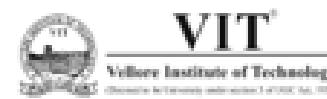
Calculate accuracy and localization-error metrics

python app.py

Launch interactive diagnostic/dashboard interface

GENERATED OUTPUTS

- Predicted (x, y) coordinates
- Localization error
- Threshold-wise accuracy
- Runtime statistics
- Per-pattern performance
- Failure-case diagnostics
- Visualization / heatmaps
- Reproducibility



EXPERIMENT SETUP & TEST DIVERSITY

TEST DIVERSITY

8 PATTERN CLASSES

+

CROSS-MAGNIFICATION

+

CONTROLLED SEM DEGRADATION

+

SPATIAL / GEOMETRIC VARIATION

+

PERIODIC HARD NEGATIVES

↓

HELD-OUT LOCALIZATION BENCHMARK

EVALUATION PROTOCOL

120 unseen pairs • 8 patterns • 15/pattern • Exact GT coordinates

STRESS COVERAGE

Noise • Blur • Scale • Rotation • Drift • Charging • Boundary cases

KEY TEST

Can the system select the correct target when several visually similar candidates exist?

EVALUATION DIMENSION	TEST DIVERSITY
Pattern Structure	FIN_ARRAY, FIN_CUT, FIN_GATE, CONTACT_ARRAY, LOCAL_INTERCONNECT, METAL_ROUTING, ACTIVE_CELL, FINFET_FULL_CELL
Magnification	100× Reference → 10× Search field relationship
Spatial Variation	Continuous target locations with exact ground-truth (x, y)
Image Degradation	Noise, blur, scale variation, rotation, drift and charging
Periodic Ambiguity	Visually similar regions and hard-negative candidates
Boundary Cases	Targets near structural/boundary regions
Ground Truth	Exact target center recorded in Search-image coordinates

THRESHOLD WISE ACCURACY, ERROR STATISTICS & RUNTIME

KEY PERFORMANCE HIGHLIGHTS

98.33%

OPERATIONAL ACCURACY < 5 PX

86.67%

HIGH PRECISION < 1 PX

0.7012 PX

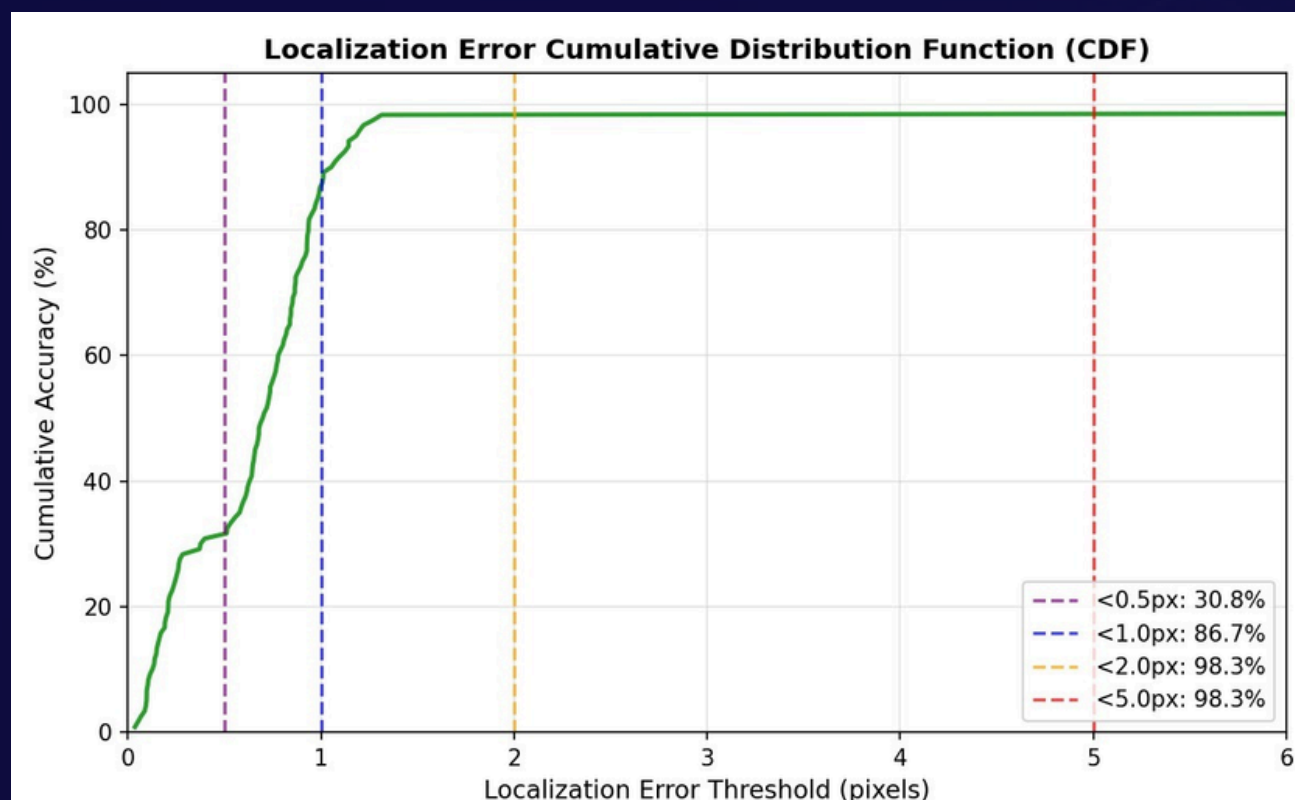
MEDIAN LOCALIZATION ERROR

672.39 MS

MEAN INFERENCE LATENCY

OVERALL PASS RATE

99.17% — 119 / 120 BENCHMARK CASES



METRIC	MEASURED VALUE	INDUSTRIAL STANDARD
Operational Accuracy (<5 px)	98.33% (118/120)	≥ 95.0%
Fine Review Accuracy (<2 px)	98.33% (118/120)	≥ 90.0%
High Precision Accuracy (<1 px)	86.67% (104/120)	≥ 75.0%
Sub-Pixel Accuracy (<0.5 px)	30.83% (37/120)	≥ 25.0%
Median Localization Error	0.7012 px	< 1.0 px
Mean Localization Error	1.5290 px	< 2.0 px
P95 Localization Error	1.1829 px	< 3.0 px
Mean Execution Latency	672.39 ms/pair	< 1000 ms

ROBUSTNESS COMPARISON AND BASELINE ABLATION

LOCALIZATION METHOD COMPARISON

APPROACH	STRENGTH	LIMITATION
Global Multi-Scale NCC	Fast candidate generation	Can produce multiple peaks in periodic layouts
Geometry-Based Disambiguation	Uses structural consistency	Activated only for ambiguous candidates
Siamese Metric Re-Ranking	Distinguishes visually similar candidates	More computationally expensive; used selectively
Fine Local Search	Improves local coordinate accuracy	Applied after candidate selection
Fourier Refinement	Provides sub-pixel peak refinement	Final localization refinement

INFERENCE PATH UTILIZATION

INFERENCE PATH	SAMPLES	PERCENTAGE
NCC Direct	114 / 120	95.00%
Geometry Verified	1 / 120	0.80%
Siamese ML Re-ranked	5 / 120	4.20%

ROBUSTNESS UNDER CONTROLLED NOISE

NOISE TIER	DOSE	DETECTOR Σ	ACCURACY <1.0 PX
LOW	3500 e-/px	0.8	87.5%
MEDIUM	2000 e-/px	1.6	83.3%
HIGH	1000 e-/px	2.4	78.1%
SEVERE	500 e-/px	3.2	75.0%

ROBUSTNESS COMPARISON & CASCADE PERFORMANCE

1. Localization Method Comparison

Classical NCC → Geometry → Siamese ML → Fine Search → Fourier Refinement

2. Inference Path Utilization

95.0% NCC Direct | 0.8% Geometry Verified | 4.2% Siamese ML

3. Robustness Under Controlled Noise

100% <5 px pass rate across all four noise tiers

FAILURE CASE, LIMITATION & LEARNING

WORST-CASE FAILURE: PERIODIC PITCH JUMP

ATTRIBUTE	VALUE
Case	PAIR_110 – P8: FINFET_FULL_CELL
Ground Truth	(269.29, 305.97)
Prediction	(270.00, 372.00)
Localization Error	66.03 px

PRIMARY LIMITATION

Highly periodic layouts combined with insufficient boundary context can make an incorrect neighbouring cell appear more similar than the true target.

KEY LEARNING

High NCC confidence alone is not sufficient when multiple periodic candidates are structurally similar.

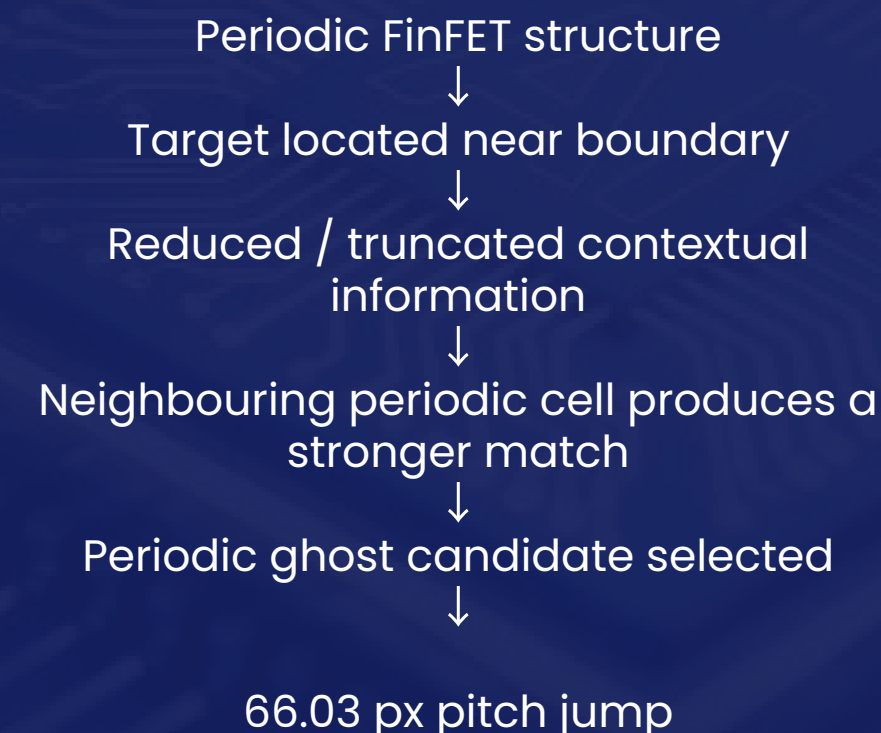
NEXT IMPROVEMENTS

- Larger macro-context around the target
- Asymmetric layout fiducials
- Peripheral boundary masking
- Improved periodic-cell disambiguation

FAILURE CHARACTERIZATION

The observed error is a structured periodic pitch jump, indicating candidate-selection ambiguity rather than a general failure of the localization pipeline.

FAILURE MECHANISM



The failure corresponds approximately to one vertical periodic cell pitch, making this a structured ambiguity rather than random localization error

CONCLUSION

FINAL BENCHMARK RESULTS — SEM

METRIC	RESULT
Operational Accuracy <5 px	98.33% (118/120)
Fine Review Accuracy <2 px	98.33% (118/120)
High Precision <1 px	86.67% (104/120)
Sub-Pixel <0.5 px	30.83% (37/120)
Median Localization Error	0.7012 px
Mean Localization Error	1.5290 px
Mean Latency	672.39 ms/pair

RGB OPTICAL BENCHMARK — EVALUATION SETUP

METRIC	RESULT
Operational Accuracy <5 px	90.00% (36/40)
Standard Tolerance <4 px	90.00% (36/40)
Fine Alignment <2 px	90.00% (36/40)
High Precision <1 px	72.50% (29/40)
Sub-Pixel <0.5 px	25.00% (10/40)
Median Localization Error	0.8233 px



- 40 augmented RGB image pairs
- Same coordinate-localization objective
- Controlled visual variations introduced through augmentation
- Evaluation at <5 px, <4 px, <2 px, <1 px and <0.5 px thresholds
- Median localization error used as the central accuracy measure



GITHUB LINK: https://github.com/DK-A/Techtonics_Drift-Sense_Wafer_Inspection_PS2



DEMO VIDEO LINK: <https://drive.google.com/file/d/1sxvBWdj1c7RfX7oeG66R8FJaaMt8o1l/view?usp=sharing>